WANG HOUTIAN

Undergraduate Physics Student · Nanyang Technological University
WA0030AN@e.ntu.edu.sg · github.com/noahwalkerror-hash · Singapore

EDUCATION

Nanyang Technological University, Singapore **Expected 2028**
Bachelor of Science in Physics
CGPA: 4.38/5.00
Transferred from Materials Science and Engineering to Physics in 2026.

RESEARCH EXPERIENCE

From the Quantum Particle to Quantum Field Theories: A Mathematical Tour **August 2026-Present**
URECA Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Developing a mathematically structured route from single-particle quantum mechanics to nonrelativistic and relativistic quantum field theories.
- The project aims to construct a dictionary between one-particle and second-quantized formulations, covering Fock spaces, creation and annihilation operators, operator-valued fields, and selected foundational aspects of Yang-Mills theory.

Memory Effects in Stochastic Thermodynamics **May 2026-Present**
Undergraduate Research Project · Supervisor: Prof. François Gay-Balmaz · NTU

- Investigating memory effects in stochastic thermodynamics through a variational formulation proposed by Prof. François Gay-Balmaz.
- Applying the framework to the full non-Markovian system, examining fluctuation-dissipation consistency, stationary behavior, and entropy production.

RESEARCH INTERESTS

Undergraduate focus: Non-Markovian stochastic dynamics; stochastic thermodynamics and nonequilibrium statistical mechanics; mathematical foundations of quantum mechanics and quantum field theory.
Future directions: Symmetry and representation theory; holography and AdS/CFT; quantum information theory.

TECHNICAL PROJECTS AND LEADERSHIP (WIP)

Illuminatio Physica **2026-Present**
Creator and Developer · illuminatio.techecho.org

- Designed and developed an interactive 3D knowledge graph for exploring physics concepts, semantic relationships, prerequisites, and learning paths.

Tech Echo Collective **2026-Present**
Founder and Developer · techecho.org

- Founded and developed a multilingual open-source platform for technical projects and long-form discussion, with GitHub-based identity, project attribution, and contributor history.